



ME 300:
Engineering Thermodynamics

Heat and Work

Heat

→ NOT A PROPERTY

→ exchange of energy to/from system from/to the surrounding,

without transfer of mass, due to a temperature difference

- occurs at boundaries

$$\Rightarrow Q = \text{heat [J]}$$

$$\dot{Q} = \text{heat transfer rate [J/s] } \sim [\text{W}]$$

$$\dot{Q}'' = \text{heat flux } \sim [\text{W/m}^2]$$

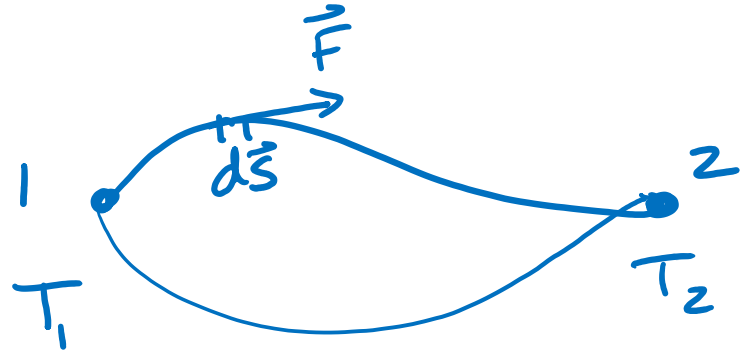
$$\Rightarrow \text{Adiabatic: } Q=0$$

Work

$\Rightarrow$ NOT A PROPERTY

$$\Rightarrow {}_1W_2 = \int \vec{F} \cdot d\vec{s}$$

$${}_1\bar{W}_2 \sim [J]$$



$$\delta W = \vec{F} \cdot d\vec{s}$$

= inexact differential

dT - exact differential $\hookrightarrow$ path dependent

$$\Rightarrow \underline{P} = \frac{W}{\text{time}} = \dot{W} = \vec{F} \cdot \frac{d\vec{s}}{dt} = \vec{F} \cdot \vec{v} \sim [J/s] \sim [W]$$



A piston compresses a cylinder of air from 10 m^3 to 5 m^3 moving at 1 m/s – would the work [5] done on the air inside be higher, lower, or the same if the piston moved at 10 m/s ?

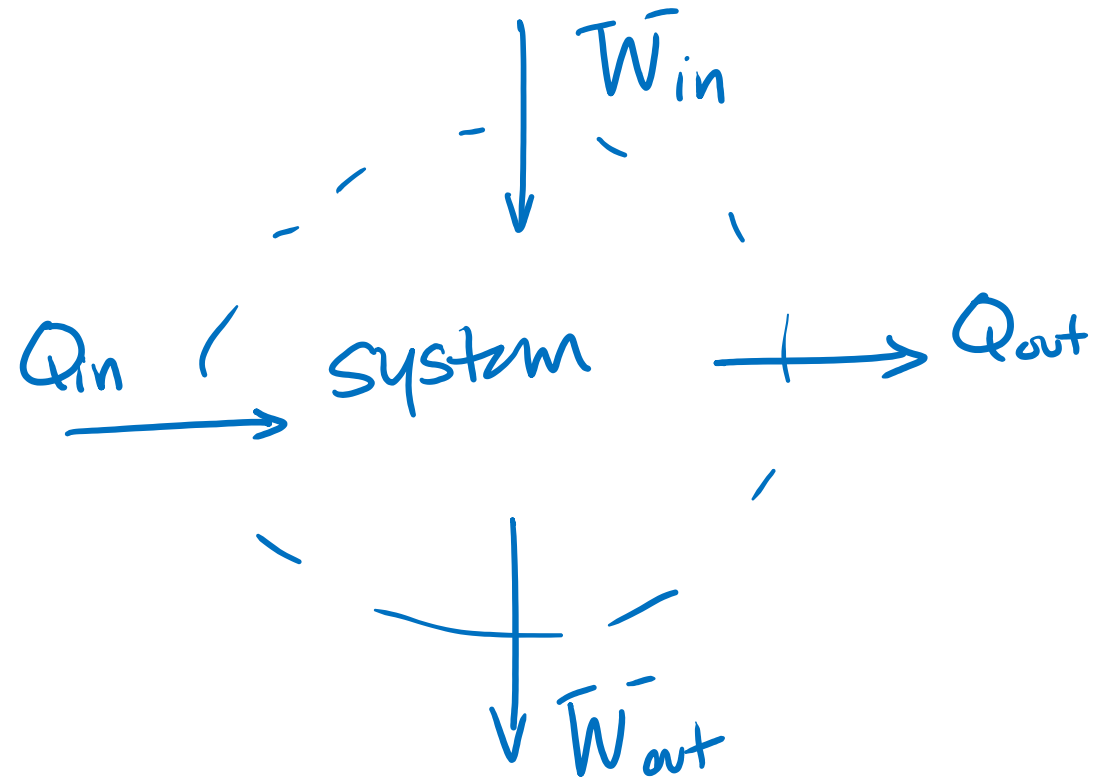
- a) Higher
- b) Lower
- c) The same



A piston compresses a cylinder of air from 10 m^3 to 5 m^3 moving at 1 m/s – would the power required be higher, lower, or the same if the piston moved at 10 m/s ?

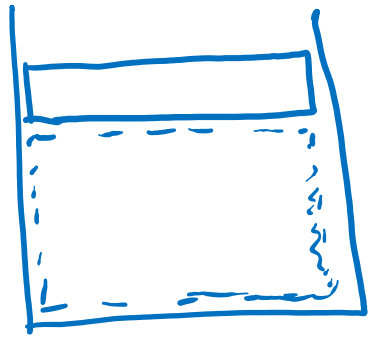
- a) Higher
- b) Lower
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Heat and work



Types of work

1. Compression/expansion work



↓ Compression
on system

↑ Expansion
by system

$$\delta W = F ds \Rightarrow P = \frac{F}{A}, A ds = dV$$

$$\delta \bar{W} = P dV$$

$$\bar{W}_2 = \int_{V_1}^{V_2} P dV$$

Compression: $dV < 0$, $\bar{W}_2 = \int_{V_1}^{V_2} P dV < 0$ work done on system: -

Expansion: $dV > 0$, $\bar{W}_2 = \int_{V_1}^{V_2} P dV > 0$ work done by system +

2. Viscous work

3. Shaft work

4. flow work